



Bulletproof Tesla

There's a bullet proof version of the Tesla Model S in the upcoming movie, The Brothers Grimsby. <http://dgit.in/TeslaBltPrf>



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world by surprise, Bezos announced on Twitter — with a video, no less! — that the New Shepard rocket managed to achieve perfect Vertical Takeoff Vertical Landing (VTVL) after reaching outer space. New Shepard crossed the Kármán Line and reached outer space at 105 km.

Bezos's announcement not only brought him a lot of attention, but also triggered a slew of defensive responses from Musk who chose to explain why Falcon 9 and New Shepard are not really comparable.

Musk is right. Blue Origin's New Shepard (NS), while impressive, is not exactly an equal competitor to SpaceX's Falcon 9 (F9). Although these two are currently the hot favourites in any reusable rocket discussion, they are two very different projects.

The instant you see images of F9 and NS side by side, you know they're different. F9 is thin and long, while NS is short and stubby. F9 is meant for launching

stage separates and begins descent. Also, since F9 aims to get to orbital velocity, its velocity at approximately 9000 kmph is thrice as high during separation as the SN's 3000 kmph. This, in addition to different orientation as well. The Falcon 9 turns parallel to the Earth's surface during first stage separation, while NS remains vertical, the way it took off.

All of this is important because each private company has different targets and as a whole are bringing space travel closer to the common man. Apart from these two, there are also other names. The third biggest is probably Richard Branson's Virgin Galactic. The SpaceShipTwo is a "spaceplane" that can fly both in air and space, and gets launched after being carried up to a specific altitude. Virgin Galactic infamously has people with tickets waiting in line to get to space on the SpaceShipTwo, but hasn't successfully managed to launch it into orbit safely. In fact, the breakup of a test craft in late 2014 causing the death of a pilot was a rude reminder to the world just how delicate and fragile spaceflight still is.

Yet another name that crops up is XCOR, whose Horizontal Takeoff Horizontal Landing (HTHL) craft, much like SpaceShipTwo, is called Lynx and is scheduled to be tested sometime next year. The British company Reaction Engines Limited meanwhile has a similar plan for Skylon, funded by the British govt. Skylon is an ambitious craft that aims to take off and land on a runway like a regular aeroplane. Sierra Nevada's Dream Chaser targets almost exactly the same mechanism. India's space agency, ISRO, is looking to develop a shuttle program as well. There are other companies that are known for their satellite making prowess, like Orbital, that both makes and launches its own satellites in its own rockets.

All of them are ambitious with ground breaking designs that will bring about radical changes in the space sector in the process of transformation to space "industry". All of the private companies in the space industry are trying to achieve one common goal: reusability.

Reusability

That's the magic word, really. In any industry, reusability is always a key component in the manufacturing process. It lowers development costs, improves accuracy, and increases safety.

Would you fly if every flight from Mumbai to New Delhi cost you Rs. 15 lakh. Then once you got there you saw them dump the plane in the ocean, before rebuilding a plane from scratch for the next flight home? This is exactly where rockets are today. The average cost of a rocket launch was somewhere between \$200 million to \$500 million. The now reusable Falcon 9, in December 2015, charged just \$54 million, and this cost will go down with every reusable launch. This is why we feel that reusable rockets are more promising than HTHL experimental crafts at the moment.

Reusability comes at a much higher initial cost though. Still, since we're not going to give up on space anytime soon, and we will need more and more satellites to enhance our communications and GPS over the decades, it's not like that research will be in vain. Reusable components go a long way towards reducing the space debris floating around.

The future

Fifty years from now, you won't just fly to space, you will get a choice of packages. You could just do a simple crossing of the Kármán line and coming back, or you could orbit the Earth a couple of times.

A century from now we hope the scenario we painted in the beginning of this article will be achievable. Hopefully, SpaceX will realise their dream, and allow our great-grandkids to choose if they want to travel and settle on Mars or Venus! Maybe probes digging around on Jupiter's and Saturn's icy moons will report signs of life back to our descendants. It's the only real frontier left, and the universe is unlimited in sheer size comparison to us. We're going to have to get off this rock eventually to survive, and we are the generation that will be remembered through history as the ones who started and supported the first initial steps to achieving just that. 🚀



It's all progress anyway right Mr. Musk?

payloads into orbit around the Earth and to the ISS. NS is meant for suborbital flights, which means the flight will not complete a full orbit around Earth, and carrying a maximum of 4 people. In keeping in line with the purpose, the two rockets have very different designs. F9 is very thin and streamlined so as to prevent drag and exit the earth's atmosphere easily to go into space. NS, on the other hand, has a thick sturdy, shorter body and is much wider in girth than the F9. Both the rockets attempt to land the first stage back. This is the stage that usually breaks away from the structure and falls back into the sea, as the rest of the components (second stage and higher) proceed further into their journey.

The height and velocity at which the stage breaks away for each rocket are also important factors to be taken into consideration. At nearly 200 km, F9 goes twice as high as NS does before the first